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HieFormer: Leveraging hyperbolic geometry to overcome the hierarchical expressiveness limitation of transformers

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ABSTRACT

3D instance segmentation requires fine-grained geometric discrimination together with an effective representation of the hierarchical organization of real-world objects and scenes. Conventional transformer architectures, however, operate predominantly in Euclidean space, whose polynomial volume growth can be poorly matched to strongly tree-like structures and can contribute to feature crowding in cluttered scenes. We propose HieFormer, whose central contribution is a unified dual-geometry design for 3D instance segmentation: hyperbolic geometry provides a hierarchy-aware space for global structural reasoning, while Euclidean geometry is retained for fine-grained local modeling, and the two are coupled within the same Transformer decoder. This design is instantiated by a learnable-curvature projection that adjusts the hyperbolic geometry during training, a hyperbolic relative position encoding (HRPE) that injects geodesic relational priors, and a Hybrid Hyperbolic and Euclidean Attention (HHEA) module that integrates hierarchical positional reasoning with discriminative Euclidean feature interactions. Extensive experiments on ScanNetv2, ScanNet200, and S3DIS show state-of-the-art or highly competitive performance. We further examine final decoder query representations under controlled structural incompleteness. When instance points are progressively removed, HieFormer exhibits a substantially more consistent ordered representation response than the Euclidean baseline, while UMAP visualization shows clearer organization across retention conditions. These findings provide representation-level evidence consistent with the intended hierarchy-aware structural modeling of HieFormer.

1. Introduction

The rapid development of 3D sensing technologies, particularly LiDAR and RGB-D cameras, has significantly increased the demand for robust perception algorithms capable of interpreting complex 3D scenes. Among these technologies, 3D instance segmentation is a critical task. It involves identifying individual objects within a scene and assigning each object a unique instance label, such that all points belonging to the same object share this label even when they belong to the same semantic class (Dai et al., 2017; Qi et al., 2017a). This capability is essential for numerous applications, including autonomous driving, where distinguishing vehicles from pedestrians is critical; robotics, where precise object manipulation relies on well-defined instance boundaries; and augmented reality, where comprehensive scene understanding enhances immersive experiences (Armeni et al., 2016; Nie et al., 2020). Despite its critical role, 3D instance segmentation remains exceptionally challenging due to the inherently irregular and unordered structure of point clouds, the computational complexity of large-scale scenes, and funda-

mentally, the necessity to **effectively** model the hierarchical organization of real-world objects.

Point clouds differ from 2D images in that they lack a canonical grid format. This poses significant challenges for the direct application of traditional convolutional neural networks (CNNs). Early methods addressed this by voxelizing point clouds into 3D grids (Maturana & Scherer, 2015). However, voxelization often incurs substantial memory consumption and tends to sacrifice fine-grained geometric details. Point-based networks, such as PointNet (Qi et al., 2017a) and PointNet++ (Qi et al., 2017b), introduced permutation-invariant architectures to process raw points directly, enabling effective semantic segmentation. However, extending these architectures to instance segmentation requires distinguishing multiple instances of the same class, a task that demands both local geometric understanding and global contextual awareness. Recent approaches, such as PointGroup (Jiang et al., 2020) and SoftGroup (Vu et al., 2022), combine semantic segmentation with clustering to predict instance masks, achieving state-of-the-art results (Dai et al., 2017). Yet, these methods often rely on complex, multi-stage pipelines with

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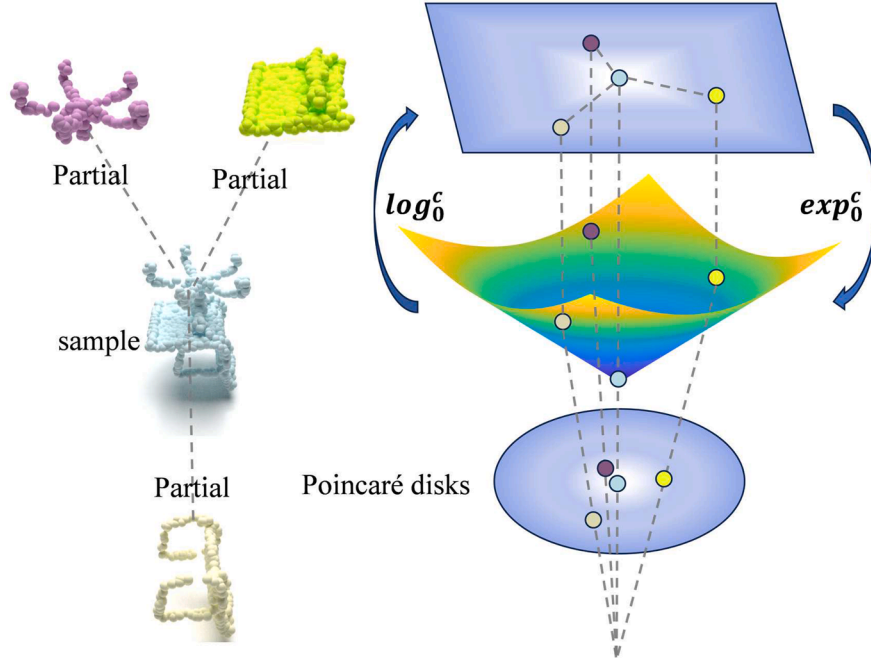


Fig. 1. Illustration of the geometric mismatch. 3D scenes exhibit an inherent hierarchical structure in which parts compose objects and objects compose scenes (e.g., legs, seat, and back forming a chair). In Euclidean space, hierarchical components can become crowded in a flat representation. Hyperbolic space provides exponentially increasing representational volume with radius, offering a geometry that is better aligned with tree-like organization and can reduce crowding among nested components.

separate branches for semantic and offset predictions, limiting their scalability and end-to-end learnability.

Recent years have witnessed the emergence of transformer-based approaches (Wu et al., 2024; Zhao et al., 2021) as a dominant paradigm in point cloud processing. Transformers, distinguished by their global modeling capacity via self-attention, have revolutionized various modalities. For point clouds, Mask3D (Schult et al., 2023) and OneFormer3D (Kolodiazhnyi et al., 2024a) use learnable queries to directly generate instance masks, bypassing complex clustering steps (Cheng et al., 2022). In practice, these methods primarily operate in Euclidean space, which may not fully align with the intrinsic tree-like hierarchy of real-world 3D objects such as a chair composed of a backrest and legs. As illustrated in Fig. 1, tree-like hierarchical structures can exhibit approximately exponential growth in the number of elements with increasing depth. Embedding these exponentially growing structures into a flat Euclidean space, which is characterized by polynomial growth, can lead to a geometric mismatch. This mismatch can cause distinct object parts to become crowded in the feature space, increasing representation distortion and degrading the model’s ability to distinguish adjacent instances with similar geometric components.

Hyperbolic geometry offers a transformative solution to this challenge, as its unique mathematical properties make it exceptionally suited for modeling hierarchical and tree-like structures (Ganea et al., 2018; Nickel & Kiela, 2017). Unlike Euclidean space, where volume grows polynomially, hyperbolic space exhibits exponential volume expansion, allowing complex hierarchical relationships to be embedded with minimal distortion in a compact manifold. The Poincaré ball model, a widely adopted representation of hyperbolic geometry, represents data in a bounded negatively curved space whose geodesic structure is well suited to tree-like organization. In learned hierarchical embeddings, radial position and geodesic distance can provide a natural representation of coarse-to-fine relationships (Tifrea et al., 2018). This property enables hyperbolic space to naturally represent the nested, tree-like structures of 3D point clouds, as illustrated in Fig. 1, empowering the network to resolve the crowding problem that limits standard Euclidean models.

To bridge this geometric gap, we propose a novel instance segmentation network called **HieFormer**. The central novelty of HieFormer lies in the *integration* of two complementary representation geometries within a unified Transformer framework, rather than in using either hyperbolic geometry or a hybrid attention operator in isolation. Hyperbolic space is assigned the role of hierarchy-aware structural reasoning, where its exponential volume growth and geodesic metric provide a natural inductive bias for nested relationships, whereas Euclidean space is retained for fine-grained local feature interactions. HieFormer realizes this dual-geometry principle through three coordinated mechanisms. First, a learnable-curvature projection maps Euclidean features to a Poincaré manifold whose curvature is optimized jointly with the network. Second, hyperbolic relative position encoding (HRPE) injects geodesic relational priors into attention. Third, Hybrid Hyperbolic and Euclidean Attention (HHEA) explicitly couples hyperbolic positional reasoning with Euclidean feature refinement. The contribution is therefore the geometry-level integration of global hierarchical reasoning and local discriminative modeling, with the individual modules serving complementary roles within this unified design.

The main contributions are summarized as follows:

- We formulate HieFormer as a unified **dual-geometry Transformer framework** for 3D instance segmentation. Its central design principle is to assign global hierarchy-aware reasoning to hyperbolic space while preserving fine-grained local modeling in Euclidean space, thereby addressing the geometric mismatch of purely Euclidean representations without discarding their local discriminative advantages.
- We instantiate the hyperbolic branch with learnable curvature and HRPE. The curvature is optimized jointly with the network to adapt the manifold geometry to the training data, while HRPE uses Poincaré geodesic relations to inject hierarchy-aware spatial priors into query refinement.
- We introduce HHEA as the integration mechanism between the two geometries. HPA captures hierarchy-aware positional relations in hy-

perbolic space, whereas ECA preserves discriminative channel interactions in Euclidean space; their fusion combines complementary structural and local information in each decoder stage.

- Extensive experiments on ScanNetv2, ScanNet200, and S3DIS, together with ablation, sensitivity, and representation analyses, demonstrate state-of-the-art or highly competitive performance. Under controlled structural incompleteness, HieFormer also exhibits a more consistent ordered response in its final decoder representations than the Euclidean baseline, providing representation-level evidence from the learned feature space.

The remainder of the paper is organized as follows: Section 2 reviews related work on 3D instance segmentation, transformer-based methods, and hyperbolic transformers. Section 3 details the HieFormer architecture, including its hyperbolic components. Section 4 presents experimental results, and Section 5 concludes with future directions.

2. Related work

2.1. 3D instance segmentation

3D instance segmentation aims to assign a unique instance label to each point in a point cloud while differentiating between distinct objects of the same semantic class. Early approaches relied heavily on hand-crafted features and geometric heuristics, such as region growing (Rabbani et al., 2006) and graph-based segmentation (Felzenszwalb & Huttenlocher, 2004). However, their dependence on predefined rules limited their robustness in complex, cluttered scenes. The advent of deep learning fundamentally transformed this field, with pioneering works such as PointNet (Qi et al., 2017a) and PointNet++ (Qi et al., 2017b) introducing permutation-invariant architectures capable of processing raw points directly, thereby laying a solid foundation for modern learning-based segmentation.

Contemporary methods can be broadly categorized into proposal-based and proposal-free approaches. Proposal-based methods, such as 3D-BoNet (Yang et al., 2019) and GSPN (Yi et al., 2019), follow a detect-then-refine strategy by first generating object proposals (e.g., bounding boxes) and subsequently refining them into instance masks. Although effective for simple objects, these methods incur high computational costs and are sensitive to the quality of initial proposals. Proposal-free methods, such as SGP (Wang et al., 2018) and ASIS (Wang et al., 2019), take a different approach by learning point-wise embeddings directly from the data, enabling points to be clustered into instances without explicit proposals. Recent high-performing methods have advanced further by combining semantic segmentation with offset-based clustering, as exemplified by PointGroup (Jiang et al., 2020) and SoftGroup (Vu et al., 2022). In a related line of work, IA-BEV (Jiao et al., 2024) proposes an instance-aware multi-camera 3D detection framework that explicitly mines category-specific structural priors to guide depth estimation and BEV feature construction, demonstrating that incorporating structural priors can significantly benefit instance-level 3D perception. Despite their promising results, these methods typically require multi-branch pipelines and rely on Euclidean distance thresholds for clustering, which often fail to capture the intricate, nested structures of complex 3D objects, leading to suboptimal segmentation boundaries.

2.2. Transformer-based methods

Transformers, originally designed for natural language processing (Vaswani et al., 2017), have become a dominant force in 3D point cloud understanding due to their capability to model long-range dependencies. Architectures like Point Transformer (Zhao et al., 2021) and Point Transformer V2 (Wu et al., 2022a) adapt self-attention mechanisms to local 3D neighborhoods, achieving robust feature extraction for semantic and instance tasks. Similarly, Stratified Transformer (Lai et al., 2022) introduced hierarchical attention strategies to process large-scale point

clouds efficiently, though often at the cost of complex preprocessing steps.

For 3D instance segmentation, recent works such as Mask3D (Schult et al., 2023) and OneFormer3D (Kolodiazhyi et al., 2024a) have adopted a set-prediction paradigm inspired by 2D MaskFormer (Cheng et al., 2022). These methods utilize Transformer decoders with learnable queries to directly predict instance masks via bipartite matching, achieving impressive accuracy on indoor datasets. Recent 3D methods also illustrate a broader design principle of assigning complementary modeling roles to different representation components. LitePT (Yue et al., 2026) employs convolutions in early high-resolution stages to capture low-level geometry and switches to attention in deeper stages to model high-level semantics and context. TG4MM (Zhang et al., 2026), although targeting 3D motion magnification rather than point-cloud segmentation, similarly decouples major spatial structure and subtle motion information through complementary space and motion triplane representations. HieFormer follows this general principle of representational complementarity but introduces it at the geometry level: hyperbolic space is used for hierarchy-aware global structural reasoning, whereas Euclidean space is retained for fine-grained discriminative interactions, and HHEA explicitly couples the two. This distinction is important because increasing contextual capacity within an entirely Euclidean representation does not alter its underlying metric geometry; for strongly tree-like organization, feature crowding can therefore remain. HieFormer complements these feature- or module-level designs by explicitly coordinating two representation geometries within the same Transformer decoder.

2.3. Hyperbolic deep learning

Hyperbolic geometry has emerged as a transformative tool for representation learning, particularly for data exhibiting latent hierarchical structures. Its defining property, exponential volume growth, allows for low-distortion embeddings of trees and scale-free networks (Ganea et al., 2018; Nickel & Kiela, 2017). In the deep learning domain, hyperbolic embeddings have proven effective for tasks such as knowledge graph reasoning (Chami et al., 2019) and hierarchical image classification (Khrulkov et al., 2020), significantly outperforming Euclidean counterparts in capturing “parent-child” relationships.

In the context of attention mechanisms, researchers have begun exploring Hyperbolic Transformers. Ganea et al. (2018) formalized operations like Möbius addition and linear layers, enabling neural networks to function within the Poincaré ball. Recent studies (Atigh et al., 2022; Li et al., 2025) have extended hyperbolic representation learning to dense prediction and structured relational modeling, further demonstrating the utility of non-Euclidean geometry for capturing complex relational structure. In 3D vision, Montanaro et al. (2022) proposed a hyperbolic encoder (HyCoRe) for point cloud classification. However, this application was limited to global shape analysis and did not address the dense, fine-grained discrimination required for instance segmentation.

To the best of our knowledge, HieFormer is the first framework to integrate hyperbolic geometry into a Transformer-based architecture specifically for 3D instance segmentation. Rather than simply operating in a non-Euclidean manifold, HieFormer bridges the gap between geometries. It leverages hyperbolic relative position encoding and a hybrid attention mechanism to explicitly model the hierarchical dependencies between object components. HieFormer combines learnable curvature with dual-space interactions to construct a geometry-aware representation tailored to dense 3D instance segmentation, achieving efficient and accurate segmentation on large-scale benchmarks.

To clearly position HieFormer relative to existing hyperbolic representation learning methods, we emphasize that its principal distinction is the *joint dual-geometry design*. HieFormer does not treat hyperbolic geometry as a wholesale replacement for Euclidean representation. Instead, learnable curvature and HRPE construct a hierarchy-aware hyperbolic branch, while HHEA couples this branch to Euclidean local feature modeling. This organization is specifically motivated by dense

Table 1

Comparison between HieFormer and prior hyperbolic representation learning methods.

Method	Geometry	Curvature	Pos. Encoding	Attention Design	Primary Task
Poincaré (Nickel & Kiela, 2017)	Hyperbolic	Fixed	N/A	N/A	Graph embedding
HGCN (Chami et al., 2019)	Hyperbolic	Learnable	Euclidean	Graph convolution	Node classification
HNN (Ganea et al., 2018)	Hyperbolic	Fixed	N/A	Hyperbolic only	Classification
HyCoRe (Montanaro et al., 2022)	Hyperbolic	Fixed	N/A	Global pooling	Point cloud classif.
HieFormer (Ours)	Hybrid	Learnable	Hyp. geodesic	Hybrid (HPA + ECA)	3D Instance Seg.

3D instance segmentation, where the representation must simultaneously separate nested structures and preserve subtle local boundaries. Table 1 summarizes the role of HieFormer relative to representative hyperbolic methods and highlights the combination of learnable geometry, geodesic positional encoding, and explicit hyperbolic and Euclidean interaction.

3. Methodology

HieFormer is built around a dual-geometry design principle: hierarchy-aware structural reasoning is performed in hyperbolic space, while fine-grained discriminative modeling is retained in Euclidean space. The central methodological contribution is the coordinated interaction of these two regimes within a single Transformer decoder. As shown in Fig. 2, this principle is instantiated by three tightly coupled components: (1) learnable-curvature projection, which establishes the hyperbolic representation space; (2) HRPE, which injects geodesic hierarchy-aware positional priors; and (3) HHEA, in which HPA and ECA jointly refine queries using hyperbolic structural relations and Euclidean local feature interactions. The components are therefore not independent additions but complementary mechanisms that implement the same dual-geometry representation strategy.

3.1. Baseline

3D Backbone. Assume the input point cloud consists of N points, and the input can be represented as $\mathbf{P} \in \mathbb{R}^{N \times 6}$, where each point is described by its RGB color values (r, g, b) and 3D coordinates (x, y, z) . In line with previous work, we voxelize the point cloud data for regular input. To extract point-wise features, we employ a U-net style backbone that incorporates submanifold sparse convolution (SSC) or standard sparse convolution (SC). The resulting feature matrix $\mathbf{P}' \in \mathbb{R}^{N \times C}$ represents the transformed features for each point. Unlike common grouping-based methods, our approach does not introduce separate semantic or offset branches, streamlining the overall process.

Superpoint pooling. To construct an end-to-end framework, we directly feed the point-wise features $\mathbf{P}' \in \mathbb{R}^{N \times C}$ into a superpoint pooling layer, which operates based on pre-computed superpoints. The superpoint pooling layer computes the superpoint features $\mathbf{S}' \in \mathbb{R}^{M \times C}$ by applying average pooling over the point-wise features within each superpoint. Without loss of generality, we assume that M superpoints are derived from the input point cloud. Notably, the superpoint pooling layer effectively downsamples the point cloud, reducing it to hundreds of superpoints. This significant reduction in data size not only alleviates the computational burden of subsequent processing stages but also enhances the representation capacity of the entire network.

3.2. Hyperbolic geometry and the Poincaré ball

Hyperbolic geometry has recently attracted significant attention in representation learning because its exponential volume growth is well matched to tree-like data (Ganea et al., 2018; Liu et al., 2019). In contrast to Euclidean space, where the volume of a ball grows polynomially with radius, hyperbolic space provides exponentially increasing capacity and is known to support low-distortion embeddings of hierarchical

structures (Nickel & Kiela, 2017). In the context of 3D point clouds, objects exhibit implicit coarse-to-fine organization; for instance, a chair is composed of parts such as legs, a backrest, and a seat (Montanaro et al., 2022). We therefore use the Poincaré ball as a hierarchy-aware representation space for structural reasoning. This geometry does not itself impose explicit ground-truth parent-child labels; rather, it provides an inductive bias whose geodesic structure is compatible with nested organization and can reduce feature crowding among structurally related components.

To fully exploit the representational capacity of hyperbolic geometry while maintaining numerical stability for gradient-based optimization, we select the Poincaré Ball model $(\mathbb{D}_c^n, g^{\mathbb{D}})$ as our embedding manifold. Here, n denotes the dimensionality, and $c > 0$ is a learnable curvature parameter, so that the manifold has constant negative curvature $-c$. Mathematically, the manifold $\mathbb{D}_c^n$ is defined as the open ball in $\mathbb{R}^n$:

$$\mathbb{D}_c^n = \{x \in \mathbb{R}^n : c\|x\|^2 < 1\}, \quad (1)$$

where $\|\cdot\|$ denotes the standard Euclidean norm. The geometry of this space is governed by the Riemannian metric tensor $g_x^{\mathbb{D}} = \lambda_x^2 g^E$, where g^E is the Euclidean metric and λ_x is the *conformal factor* defined as:

$$\lambda_x = \frac{2}{1 - c\|x\|^2}. \quad (2)$$

This conformal factor λ_x approaches infinity as x moves towards the boundary ($c\|x\|^2 \rightarrow 1$), strongly magnifying local metric distances near the boundary. More generally, hyperbolic volume grows exponentially with geodesic radius, providing increasing representational capacity for hierarchical structures. This boundary behavior provides additional geometric capacity that can facilitate the separation of representations associated with different structural scales.

Since standard vector addition is not closed within the Poincaré ball, we utilize Möbius addition to generalize translation. For any two points $x, y \in \mathbb{D}_c^n$, the operation is defined as:

$$x \oplus_c y = \frac{(1 + 2c\langle x, y \rangle + c\|y\|^2)x + (1 - c\|x\|^2)y}{1 + 2c\langle x, y \rangle + c^2\|x\|^2\|y\|^2}, \quad (3)$$

where $\langle \cdot, \cdot \rangle$ denotes the standard Euclidean inner product. This operation preserves the hyperbolic structure and is essential for implementing position biases.

The distance between two points x, y in hyperbolic space is measured along the geodesic (shortest path), given by:

$$d_{\mathbb{D}_c^n}(x, y) = \frac{2}{\sqrt{c}} \operatorname{arctanh}(\sqrt{c}\|x \oplus_c y\|). \quad (4)$$

For tree-like embeddings, radial position in the Poincaré ball can encode hierarchy depth, with higher-level representations typically occupying smaller radii and finer-grained representations occupying larger radii. The geodesic metric therefore provides a geometry compatible with hierarchy-aware separation. In this sense, the hyperbolic distance acts as a hierarchy-aware similarity metric. Compared with Euclidean geometry, the Poincaré metric increasingly magnifies separations as representations approach the boundary, while the available hyperbolic volume grows exponentially with geodesic radius. This geodesic distance is used in the Hyperbolic Positional Attention to compute attention weights (Section 3.5); the same geodesic structure also underlies the Hyperbolic Relative Position Encoding (Section 3.4), which builds

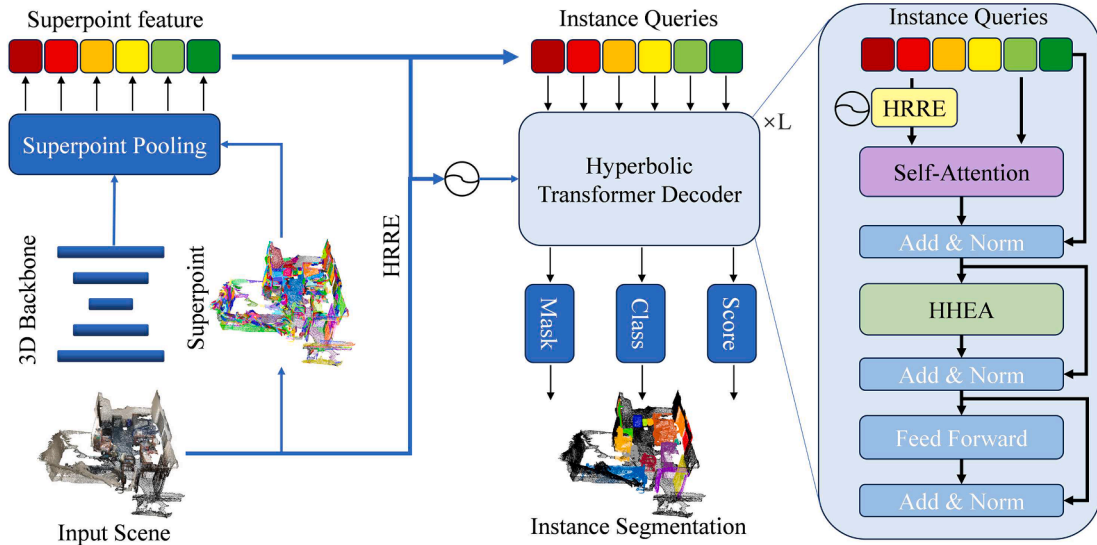


Fig. 2. The overall architecture of **HieFormer**. The hierarchical positional relationship of different instance queries is encoded through hyperbolic relative position encoding (HRPE), which serves as geometric prior information to dynamically adjust the self-attention weights. In the hyperbolic self-attention mechanism, an attention matrix incorporating hierarchical information is constructed via a hyperbolic linear layer. Subsequently, query differentiation and refinement are carried out through the Hybrid Hyperbolic and Euclidean Attention (HHEA) module.

hierarchy-aware positional priors from the corresponding Möbius difference in the Poincaré ball.

To integrate hyperbolic operations into our neural network pipeline, we need a mechanism to transition between the Euclidean embedding space (output by the backbone) and the hyperbolic manifold. We employ the Exponential map to project features from the tangent space $T_x \mathbb{D}_c^n$ to the manifold $\mathbb{D}_c^n$, and the Logarithmic map for the inverse. Assuming the tangent space is located at the origin $\mathbf{0}$ (where $T_0 \mathbb{D}_c^n \cong \mathbb{R}^n$), these differentiable bijective operators are defined as:

$$\exp_0^c(v) = \tanh(\sqrt{c}\|v\|) \frac{v}{\sqrt{c}\|v\|}, \quad (5)$$

$$\log_0^c(y) = \operatorname{arctanh}(\sqrt{c}\|y\|) \frac{y}{\sqrt{c}\|y\|}, \quad (6)$$

where $v \in T_0 \mathbb{D}_c^n$ and $y \in \mathbb{D}_c^n$. In our framework, $\exp_0^c$ is used to lift the Euclidean features into the Poincaré ball to perform geometric reasoning, while $\log_0^c$ maps the processed features back to the tangent space for compatibility with standard linear layers.

3.3. Adaptive curvature projection & hyperbolic linear layer

Standard neural networks rely on Euclidean linear transformations, which assume zero curvature, whereas the hierarchy-aware branch of HieFormer operates on a negatively curved manifold. Rather than manually fixing this curvature as a hyperparameter, we make its magnitude c learnable and optimize it jointly with the network. Specifically, c is constrained to remain positive through a softplus reparameterization and is updated by standard back-propagation without a separate optimization stage. We refer to this mechanism as **Adaptive Curvature Projection**: “adaptive” denotes adaptation during training to the statistics and hierarchical structure of the training data. In the current implementation, c is a shared learned parameter rather than a scene-conditioned prediction; this distinction is revisited in Section 4.7. The learnable parameterization removes the need to select a fixed curvature a priori while preserving a stable common manifold for HRPE and HHEA.

Formulation. Directly performing matrix multiplication on the hyperbolic manifold is computationally expensive and numerically unstable. To balance efficiency with geometric fidelity, we formulate the hyperbolic linear layer by performing the linear transformation in the tan-

gent space (approximated as Euclidean space at the origin) and applying the bias in the hyperbolic space. Formally, given an input feature x (typically in the tangent space from the backbone), the transformation is defined as:

$$y = \log_0^c(\exp_0^c(Wx) \oplus_c b), \quad (7)$$

where W is the weight matrix, b is the hyperbolic bias, and $\oplus_c$ denotes Möbius addition.

Initialization. Proper initialization is critical for convergence in hyperbolic networks. Unlike Euclidean layers where W is initialized via Kaiming uniform (He et al., 2015), we initialize W using an orthogonal distribution (Saxe et al., 2013) to act as an approximate isometry in the tangent space. The bias b is initialized to zero.

Geometric Intuition of the Bias. The role of the bias b differs fundamentally between geometries. In Euclidean space, adding b acts as a pure translation, which preserves relative distances (isometry). The spatial arrangement of points remains rigid regardless of b ’s magnitude. In contrast, the hyperbolic bias addition $\oplus_c b$ acts as a Möbius translation. As illustrated in Fig. 1, as the magnitude of b increases, it pushes points towards the boundary of the Poincaré ball. Due to the conformal factor $\lambda_x = \frac{2}{1-c\|x\|^2}$, the underlying space expands exponentially near the boundary. This property is pivotal for modeling hierarchies: it allows the network to utilize the “bias” to push finer-grained features (leaves of the tree) towards the boundary where the available volume is exponentially larger, thereby resolving the crowding problem inherent in Euclidean embeddings. This variation in point distribution highlights the unique capability of our layer to model exponentially growing structures with minimal distortion.

3.4. Hyperbolic relative position encoding

In Transformer decoders, self-attention refines instance queries by aggregating contextual information. Conventional Euclidean relative position encoding primarily describes coordinate offsets and does not explicitly provide a geometry-aware prior for nested or coarse-to-fine organization. Previous methods such as MAFT (Lai et al., 2023) model relative relations between queries and superpoints in Euclidean coordinates. In contrast, we introduce HRPE, which represents these relations using the geodesic structure of the Poincaré ball and provides a

hierarchy-aware geometric bias for query refinement. This design encourages the decoder to capture coarse-to-fine structural relationships without requiring explicit parent and child supervision.

For the position queries $\hat{Q}_i^p \in \mathbb{R}^{n \times 3}$, and the global positions $P \in \mathbb{R}^{N \times 3}$, we first project their positions to the Poincaré ball:

$$h(\hat{Q}_i^p) = \exp_0^c(\hat{Q}_i^p), \quad h(P) = \exp_0^c(P). \quad (8)$$

Here, c is the shared learnable curvature parameter optimized jointly with the network.

First, we compute the hyperbolic relative position $\mathbf{r} \in \mathbb{R}^{n \times N \times 3}$ between each position query $\hat{Q}_i^p$ and the global positions P on the manifold. Whereas the Euclidean relative position encoding of MAFT (Lai et al., 2023) uses the coordinate offset $P - \hat{Q}_i^p$, we replace it by its hyperbolic counterpart, namely the Möbius difference mapped back to the tangent space:

$$\mathbf{r} = \log_0^c((-h(\hat{Q}_i^p)) \oplus_c h(P)), \quad \hat{\mathbf{r}} = \left\lfloor \frac{\mathbf{r}}{s} \right\rfloor + \frac{L}{2}, \quad (9)$$

where $(-h(\hat{Q}_i^p)) \oplus_c h(P)$ is the Möbius difference (the hyperbolic analogue of the offset $P - \hat{Q}_i^p$), and $\log_0^c(\cdot)$ maps it back to the tangent space so that every query-position pair yields a three-dimensional signed relative coordinate distance whose Euclidean norm is proportional to the geodesic distance of Eq. (4); with the convention adopted in Eq. (4), $\|\mathbf{r}\| = \frac{1}{2} d_{\mathbb{D}_c^n}(h(\hat{Q}_i^p), h(P))$, and this constant factor is absorbed by the quantization step s in $\lfloor \mathbf{r}/s \rfloor$. Here s denotes the quantization size and L denotes the length of the relative-position encoding table. Because these relative coordinates are signed, we add $\frac{L}{2}$ to shift the quantized indices to be non-negative, and clip them to $[0, L-1]$.

Then, we use the discrete relative positions $\hat{\mathbf{r}}$ as indices to look up the corresponding position encoding tables $t \in \mathbb{R}^{3 \times L \times d}$. Formally, the relative position encoding $\mathbf{pos} \in \mathbb{R}^{n \times N \times d}$ is yielded as:

$$\mathbf{f}^{pos} = \mathbf{t}[0, \hat{\mathbf{r}}_x] \oplus_c \mathbf{t}[1, \hat{\mathbf{r}}_y] \oplus_c \mathbf{t}[2, \hat{\mathbf{r}}_z], \quad (10)$$

where $\oplus_c$ denotes Möbius addition. The resulting position encoding $\mathbf{f}^{pos}$ represents learnable query-position relationships in the hyperbolic geometry. Its parameters are optimized end-to-end from the training data, allowing the positional prior to capture recurring coarse-to-fine structural patterns without requiring explicit hierarchical annotations.

To ensure the consistency of spatial features, we need to project the hyperbolic spatial features back to the original space through logarithmic projection. Further, the relative position encoding $\mathbf{f}^{pos}$ performs dot product with the query features $f_q \in \mathbb{R}^{n \times d}$ or key features $f_k \in \mathbb{R}^{N \times d}$ in the self-attention, which is formulated as:

$$\mathbf{pos_bias}_{i,j} = \log_0^c(\mathbf{f}_{i,j}^{pos}) \cdot \mathbf{f}_i^q + \log_0^c(\mathbf{f}_{i,j}^{pos}) \cdot \mathbf{f}_j^k, \quad (11)$$

where $\mathbf{pos_bias} \in \mathbb{R}^{n \times N}$ is the positional bias. It is then added to the self-attention weights, followed by the softmax function, as shown in Fig. 3.

In summary, HRPE can be viewed as a hierarchy-aware extension of conventional relative position encoding. Unlike Euclidean relative position encoding, which directly measures coordinate offsets, HRPE measures the relative relationship between a query and the global positions through the Möbius difference in the Poincaré ball in Eq. (9), whose magnitude is proportional to the geodesic distance, quantizes the resulting three hyperbolic relative coordinates into discrete indices, and retrieves learnable encodings that are composed through Möbius addition in Eq. (10). The resulting representations therefore encode both spatial proximity and implicit hierarchical information, and injecting them into self-attention guides the decoder to aggregate information from structurally related regions while suppressing irrelevant interactions.

3.5. Hybrid hyperbolic and euclidean attention

A central motivation of HieFormer is that accurate 3D instance segmentation requires reasoning in two complementary geometric regimes. Hyperbolic space provides a natural inductive bias for global structural

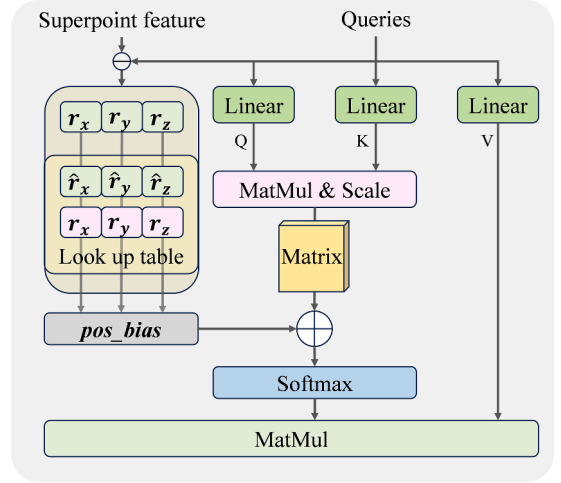


Fig. 3. The hyperbolic relative position encoding is used to introduce additional hyperbolic relative position information for the self-attention mechanism to improve the model's perception of the location information of different instances.

organization because geodesic distances can expand nested relations without forcing all components into a flat representation. Euclidean space, by contrast, remains well suited to fine-grained local feature interactions and boundary-sensitive discrimination. This perspective is related to the broader complementary-representation principle adopted by recent 3D methods (Yue et al., 2026; Zhang et al., 2026), but HieFormer introduces complementarity at the underlying representation geometry rather than only through different feature modules or processing stages.

To realize this geometry-level complementarity, we propose a **Hybrid Hyperbolic and Euclidean Attention (HHEA)** mechanism with two coordinated branches: (1) **Hyperbolic Positional Attention (HPA)**, which uses hyperbolic geodesic relations to emphasize hierarchy-aware structural compatibility between queries and superpoints; and (2) **Euclidean Channel Attention (ECA)**, which preserves discriminative channel-wise interactions in Euclidean space. Their fused output therefore combines global structural organization with local feature detail. Importantly, HHEA is not presented as an isolated attention replacement; it is the mechanism that operationalizes the central dual-geometry design of HieFormer.

Hyperbolic Positional Attention (HPA). The HPA branch performs cross-attention from each instance query to the superpoints on the manifold, using geodesic distances to prioritize superpoints that are hierarchically related to the query. Given superpoint features $\mathbf{S} \in \mathbb{D}_c^n$ (after HRPE) and a query $Q^i \in \mathbb{D}_c^n$, we obtain the query vector from Q^i and the key/value vectors from the superpoints through an on-manifold hyperbolic transformation, i.e., the hyperbolic linear layer of Eq. (7) without the final logarithmic map, so that q , k_j , and v_j remain in the Poincaré ball and the geodesic distance is well defined:

$$q = \exp_0^c(W_q Q^i) \oplus_c b_q, \quad (12)$$

$$k_j = \exp_0^c(W_k S_j) \oplus_c b_k, \quad (13)$$

$$v_j = \exp_0^c(W_v S_j) \oplus_c b_v, \quad j = 1, \dots, M, \quad (14)$$

where $W_q, W_k, W_v \in \mathbb{R}^{n \times n}$ and $b_q, b_k, b_v \in \mathbb{D}_c^n$ are trainable parameters, initialized orthogonally and with zeros respectively, and M is the number of superpoints.

The attention weights are computed directly from the hyperbolic geodesic distances between the query and the superpoints:

$$a_j = \frac{\exp(-d_{\mathbb{D}_c^n}(q, k_j)/\tau)}{\sum_{m=1}^M \exp(-d_{\mathbb{D}_c^n}(q, k_m)/\tau)}, \quad (15)$$

where $\tau = 0.1$ controls the sharpness of the distance-based attention distribution. A smaller geodesic distance therefore produces a larger at-

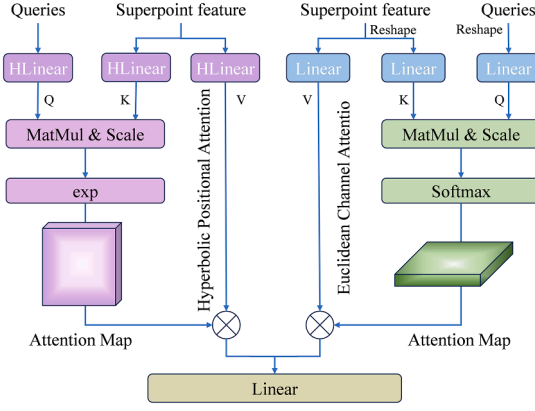


Fig. 4. HPA leverages hyperbolic distances in the Poincaré ball to capture hierarchical positional relationships between queries and superpoints, while ECA recalibrates channel-wise feature importance in Euclidean space to enhance fine-grained feature interactions. This hybrid approach refines instance queries by balancing positional hierarchy and channel-wise context, improving instance segmentation accuracy.

tention weight. The normalized scalar weights $\{a_j\}_{j=1}^M$ are subsequently used for Möbius scalar multiplication when aggregating the superpoint values.

The output of the query is then aggregated over the superpoint values through Möbius operations:

$$Q_h^l = \bigoplus_{j=1}^M (a_j \otimes_c v_j) \in \mathbb{D}_c^n, \quad (16)$$

where $\otimes_c$ denotes Möbius scalar multiplication and $\bigoplus$ denotes left-folded Möbius addition.

In this way, HPA emphasizes hierarchy-aware structural compatibility between queries and superpoints in the hyperbolic representation space. Unlike conventional dot-product attention, HPA measures query-key compatibility through the geodesic distance $d_{\mathbb{D}_c^n}(\cdot, \cdot)$. As defined in Eq. (15), smaller geodesic distances result in larger attention weights, allowing structurally related query and superpoint representations to contribute more strongly to the query update.

Euclidean Channel Attention (ECA). As shown in Fig. 4, ECA recalibrates channel-wise feature importance in Euclidean space, enhancing fine-grained feature interactions across superpoints. For each query Q_i^l and the Euclidean superpoint features S_j^e , we derive the query vector from the query and the key/value vectors from the superpoints using standard linear transformations:

$$q_i^c = W_q^e Q_i^l, \quad k_j^c = W_k^e S_j^e, \quad v_j^c = W_v^e S_j^e, \quad j = 1, \dots, M, \quad (17)$$

where W_q^e, W_k^e, W_v^e are trainable parameters and M is the number of superpoints. The attention weights are obtained by a scaled dot product followed by a softmax over the superpoints:

$$\text{Attention}^e(q_i^c, k_j^c) = \text{softmax}_j \left(\frac{(q_i^c)^T k_j^c}{\sqrt{D_c}} \right), \quad (18)$$

where D_c is the channel dimension, which serves as the scaling factor of the dot-product attention. The ECA output for query i is then:

$$Q_c^l = \sum_{j=1}^M \text{Attention}^e(q_i^c, k_j^c) v_j^c. \quad (19)$$

ECA enhances the query's channel-wise features by emphasizing channels that are most relevant to the scene's context, improving fine-grained discrimination for instance segmentation.

Fusion Mechanism. To synthesize the structural and semantic representations, we project the hyperbolic output Q_h^l to the tangent space and concatenate it with the Euclidean output Q_c^l :

$$Q_{fusion}^l = [Q_c^l, \log_0^c(Q_h^l)]. \quad (20)$$

The concatenated vector is processed by a linear layer to produce the final output in the tangent space:

$$Q_{out}^l = W_f Q_{fusion}^l + b_f, \quad (21)$$

where W_f and b_f are learnable parameters. The hybrid HPA and ECA mechanism ensures that queries are refined with both hierarchical positional information and channel-wise feature importance, reducing noise and improving mask accuracy.

In summary, HieFormer leverages both Euclidean and hyperbolic geometries to model hierarchical relationships in 3D point clouds for instance segmentation. The sparse 3D U-Net and superpoint pooling provide efficient feature extraction, while the hyperbolic linear transformation, learnable relative relationship encoding with quantized geodesic distances, and hybrid hyperbolic positional and Euclidean channel attention mechanisms operate across both spaces to capture tree-like structures and fine-grained feature interactions, enhancing the accuracy of instance mask predictions.

4. Experiments

4.1. Datasets

The experiments are performed on ScanNetv2 (Dai et al., 2017), ScanNet200 (Rozenberszki et al., 2022) and S3DIS (Armeni et al., 2016) datasets. The ScanNetv2 dataset has 1613 indoor scenes, of which 1201 are used for training, 312 for validation and 100 for testing. The dataset comprises 20 semantic categories, among which 18 object categories are used for instance-level evaluation. We report results on both validation and hidden test splits. ScanNet200 extends the original ScanNet semantic annotation with fine-grained, long-tailed categories, resulting in 198 instance categories and 2 additional semantic classes. The training, validation, and testing splits are similar to the original ScanNetv2 dataset. The S3DIS dataset contains 271 rooms in 6 areas of three buildings, and 13 semantic categories are annotated. Following previous works, the scenes in Area 5 are used for validation and the others are for training.

4.2. Metrics

Task-mean average precision (mAP) is utilized as the common evaluation metric for instance segmentation, which averages the scores with IoU thresholds set from 50% to 95%, with a step size of 5%. Specifically, mAP_{50} and mAP_{25} denote the scores with IoU thresholds of 50% and 25%, respectively. We report mAP, mAP_{50} and mAP_{25} on all datasets and we also utilized mean precision (mPrec) and mean recall (mRec) in our analysis of the S3DIS dataset.

4.3. Implementation details

We integrated the hyperbolic components in HieFormer into four different architectures (SPFormer, Mask3D, MAFT, and OneFormer3D) while retaining all hyperparameters from their original implementations. The approach for initializing queries and the backbone network configurations remain identical to those used in the respective baseline methods. For SPFormer and MAFT, the query count is set to 400 for both ScanNetv2 and S3DIS, whereas ScanNet200 requires 800 queries. In contrast, OneFormer3D dynamically adjusts the number of queries based on the superpoint distribution. Meanwhile, Mask3D employs 150 queries for ScanNetv2 and S3DIS. Regarding voxel resolution, ScanNetv2 and ScanNet200 experiments use a 2cm grid, while S3DIS adopts a slightly larger 5cm resolution due to its broader spatial coverage. Superpoint generation varies across datasets: ScanNetv2 and ScanNet200 leverage graph-based clustering followed by pooling, S3DIS follows the methodology outlined in Chibane et al. (2022).

Throughout all experiments, HieFormer is integrated into the four frameworks listed above without changing their original backbone networks, query initialization strategies, or training hyperparameters, so

that the reported gains can be attributed to the proposed hyperbolic components rather than to backbone modifications. Unless otherwise stated, the baseline numbers reported in our comparison tables are taken directly from the corresponding original publications or official benchmark leaderboards. For methods with publicly available implementations, we follow their official protocols and evaluation settings to ensure a fair comparison. Unless otherwise specified, the results reported as HieFormer (Ours) in the main comparison tables adopt OneFormer3D as the default backbone, whereas the per-backbone plug-and-play results obtained by integrating HieFormer into SPFormer, Mask3D, MAFT, and OneFormer3D are summarized separately in Tables 6 and 7.

4.4. Experimental results

Experiment Analysis on ScanNetv2 Test set. As shown in Table 2, HieFormer achieves the highest overall performance on the ScanNetv2 hidden test set, reaching 58.5% mAP, 80.1% mAP₅₀, and 88.6% mAP₂₅. These results represent consistent improvements over recent strong baselines, including QueryFormer and MSTA3D, indicating that incorporating hyperbolic geometric modeling contributes measurable benefits to 3D instance segmentation. Across the 18 instance categories, HieFormer produces the best results on 7 classes and competitive performance on most remaining ones. Notably, categories with strong structural hierarchy, such as bkshf and cabinet, show clear gains, suggesting that hyperbolic relative encoding and learnable curvature help preserve hierarchical relationships that Euclidean-only models tend to distort. Performance remains stable on rigid layout elements (e.g., bath, bed, toilet) where local geometric consistency is crucial. Although some competing models slightly outperform our method on isolated classes, HieFormer maintains competitive performance across a broad range of categories, suggesting consistent behavior across diverse object shapes and spatial arrangements. Overall, the results demonstrate that combining hyperbolic and Euclidean representations offers practical advantages in complex indoor scenes, improving both global structural reasoning and fine-grained instance discrimination.

Experiment Analysis on ScanNetv2 Validation Set. Table 3 reports the quantitative results on the ScanNetv2 validation set, while the corresponding qualitative comparison is shown in Fig. 5. HieFormer achieves 59.3% mAP, 77.8% mAP₅₀, and 86.5% mAP₂₅, delivering competitive performance across all metrics. Compared with strong recent baselines such as MAFT and MSTA3D, our method shows consistent improvements, particularly at higher IoU thresholds (mAP and mAP₅₀), where precise instance boundaries and reliable hierarchical reasoning are essential. The gains over MAFT (+1.0% mAP) and MSTA3D (+0.9% mAP) indicate that the proposed hyperbolic modeling provides benefits beyond standard Euclidean attention and multi-scale refinement strategies. Furthermore, HieFormer surpasses the recent transformer-based QueryFormer by +2.8% mAP, demonstrating stronger representational ability for complex indoor scenes. While EASE achieves a slightly higher mAP, it does not report mAP₂₅ and employs additional edge-aware refinements. In contrast, HieFormer maintains balanced improvements across all IoU thresholds, suggesting robust performance in both global reasoning and fine-grained mask prediction. Compared with its default backbone OneFormer3D, reproduced here under the same protocol, HieFormer raises mAP₅₀ and mAP₂₅ to the best values among all compared methods and is second only to the edge-refined EASE in mAP. We further note that the magnitude of the improvement varies with both the backbone and the evaluation split. The largest gain, +2.8% mAP, is observed with SPFormer, whereas the stronger OneFormer3D baseline exhibits smaller but consistent improvements. The different gain magnitudes across validation and hidden test splits may reflect differences in baseline strength and data composition, and we therefore do not attribute this variation to a single factor. Overall, these results confirm that integrating hyperbolic geometry with a hybrid attention mecha-

nism leads to more discriminative and structure-aware instance segmentation.

Experiment Analysis on ScanNet200. Table 4 presents the results on the ScanNet200 validation set, where HieFormer achieves state-of-the-art performance with 27.1% mAP, surpassing all compared methods including MSTA3D (26.2%) and SPFormer (25.2%). The model also attains the highest mAP₅₀ of 35.7% and mAP₂₅ of 40.9%, indicating robust instance segmentation across different overlap criteria. These results indicate that HieFormer's hyperbolic modeling provides consistent benefits across different IoU criteria, particularly in complex indoor scenes with diverse object layouts.

Experiment Analysis on S3DIS. As shown in Table 5, HieFormer achieves highly competitive results on the S3DIS Area 5 benchmark, obtaining 72.5% mAP₅₀, 81.2% mPrec, and 71.5% mRec. The corresponding qualitative comparison is shown in Fig. 6. These results represent either the best or second-best performance across all three metrics, outperforming these baselines on most reported metrics. Notably, HieFormer reaches the highest precision score (81.2%), indicating that the model produces cleaner and more reliable instance predictions with fewer false positives. At the same time, its recall (71.5%) remains among the top performers, demonstrating robust instance coverage even in cluttered indoor scenes. The improvements align closely with the motivation of our method. S3DIS contains large, open indoor spaces in which large structural regions, furniture-scale objects, and smaller local components coexist, creating substantial variation in spatial scale and structural organization. Purely Euclidean representations may be less well matched to such coarse-to-fine structural organization when strongly tree-like relationships are present. In contrast, HieFormer's hyperbolic modeling effectively preserves such hierarchical relationships, enabling sharper instance boundaries and more consistent region grouping. This is reflected in the higher precision compared with prior state-of-the-art models like Mask3D and QueryFormer. Moreover, HieFormer's balanced gains across precision and recall highlight the benefits of hybrid geometric reasoning: the hyperbolic branch enhances global structure alignment, while the Euclidean branch refines local geometric details. This synergy contributes to the stable improvement across all metrics and demonstrates the practical value of introducing hyperbolic geometry into transformer-based 3D perception pipelines.

4.5. Ablation study

Combine with different backbones. We integrate HieFormer's geometric modules into multiple backbones to validate its plug-and-play capability. The ablation studies systematically validate HieFormer's compatibility with diverse transformer-based backbones across datasets.

Table 4 presents the results on the ScanNet200 validation set, a challenging long-tailed benchmark with 198 instance-level categories. HieFormer achieves the best performance among the compared methods across all metrics, reaching 27.1% mAP, 35.7% mAP₅₀, and 40.9% mAP₂₅. Compared with MSTA3D, our method improves mAP by +0.9% and mAP₅₀ by +0.5%, indicating consistent gains under fine-grained category variation. The gain over SPFormer is +1.9% mAP, further supporting the benefit of the proposed geometry-aware modeling. Similarly, on S3DIS Area 5 (Table 5), HieFormer delivers the best mAP₅₀ (72.5%) while achieving the highest precision (81.2%) and second-highest recall (71.5%). These results indicate that the proposed hybrid hyperbolic and Euclidean design balances instance boundary precision and overall instance coverage in complex indoor spaces. Together, the cross-dataset gains support the robustness of the proposed dual-geometry design across different indoor benchmarks.

Ablation study of Curvature. Fig. 7 reveals that curvature affects segmentation accuracy in a non-linear manner. Performance improves steadily from $c = 0.01$ to $c = 0.10$, indicating that a moderate curvature magnitude can alleviate feature crowding while retaining useful struc-

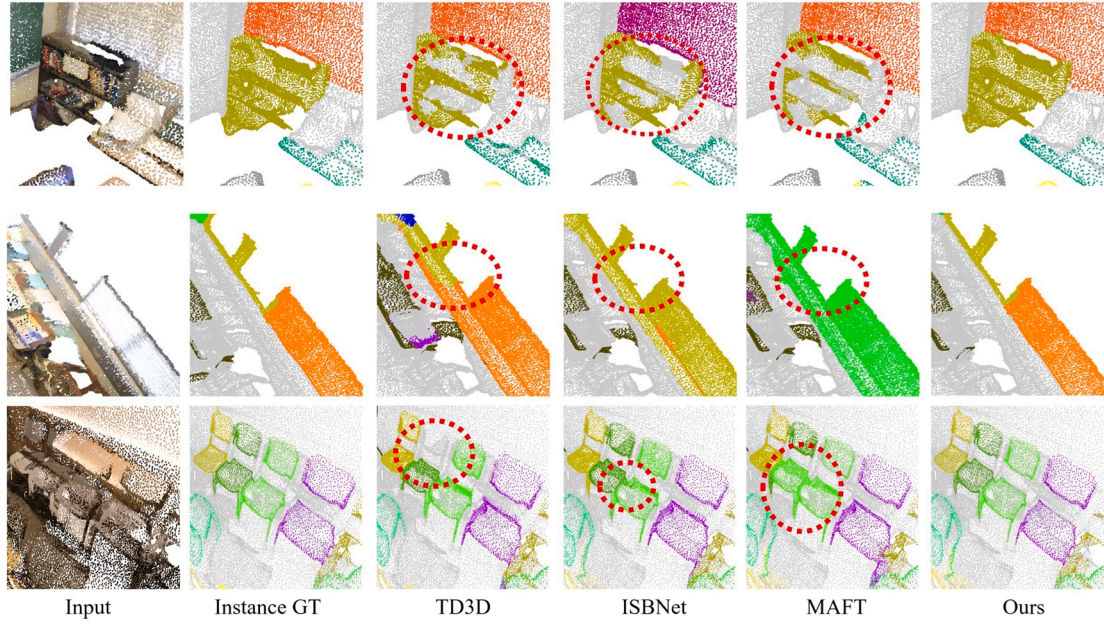


Fig. 5. Qualitative comparison of TD3D, ISBNet, MAFT, and ours on ScanNetv2 validation set.

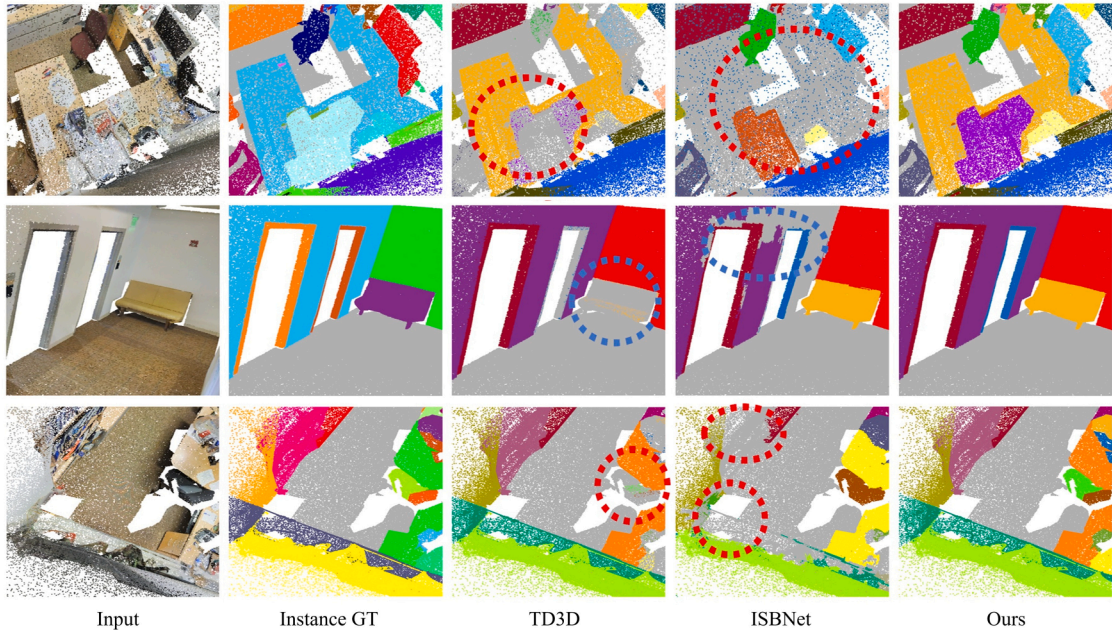


Fig. 6. Qualitative comparison of TD3D, ISBNet, and ours on S3DIS validation set.

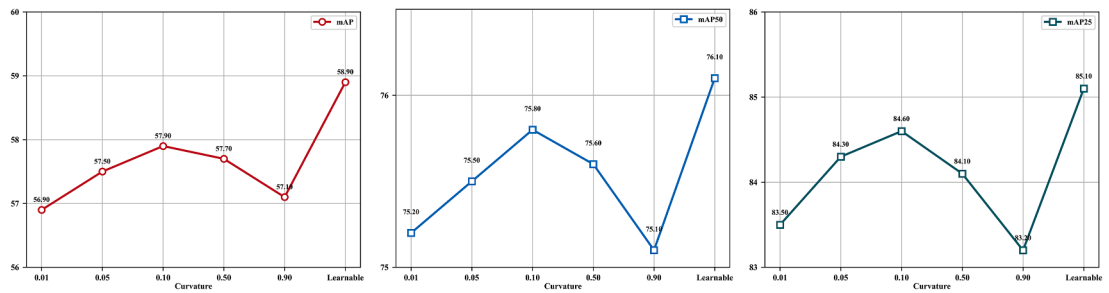


Fig. 7. Ablation study of curvature on ScanNetv2.

Table 2
Performance comparisons of 3D instance segmentation on ScanNetv2 hidden test set in terms of the mean average precision and mAP₂₅ scores for each class. Bold indicates the best option, and underline indicates the second-best option.

Methods	mAP	mAP ₅₀	mAP ₂₅	bath	bed	bksht	cabinet	chair	counter	curtain	desk	door	other	picture	fridge	s. cur.	sink	sofa	table	toilet	window
DyCo3D (He et al., 2021)	39.5	64.1	76.1	100	93.5	89.3	75.2	86.3	60.0	58.8	74.2	64.1	63.3	54.6	55.0	85.7	78.9	85.3	76.2	98.7	69.9
SSTNet (Liang et al., 2021)	50.6	78.9	69.8	100	84.0	88.8	71.7	83.5	71.7	68.4	62.7	72.4	65.2	72.7	60.0	100	91.2	82.2	75.7	100	69.1
HAIS (Chen et al., 2021)	45.7	69.9	80.3	100	99.4	82.0	75.9	85.5	55.4	88.2	82.7	61.5	67.6	63.8	64.6	100	91.2	79.7	76.7	99.4	72.6
DKNet (Wu et al., 2022b)	53.2	71.8	81.5	100	93.0	84.4	76.5	91.5	53.4	80.5	80.5	80.7	65.4	76.3	65.0	100	79.4	88.1	76.6	100	75.8
PBNet (Zhao et al., 2023)	57.3	74.7	82.5	100	96.3	83.7	84.3	86.5	<u>82.2</u>	64.7	87.8	73.3	63.9	68.3	65.0	100	85.3	87.0	82.0	100	74.4
TD3D (Kolodiaznyi et al., 2024b)	48.9	75.1	87.5	100	97.6	87.7	78.3	97.0	88.9	82.8	94.5	80.3	71.3	72.0	70.9	100	93.6	93.4	87.3	100	79.1
SoftGroup (Vu et al., 2022)	50.4	76.1	86.5	100	96.9	86.0	86.0	91.3	55.8	89.9	91.1	76.0	82.8	73.6	80.2	98.1	91.9	87.5	87.7	100	82.0
ISNet (Ngo et al., 2023)	55.9	75.7	83.5	100	95.0	73.1	81.9	91.8	79.0	74.0	85.1	83.1	66.1	74.2	65.0	100	93.7	81.4	83.6	100	76.5
SPFormer (Sun et al., 2023)	54.9	77.0	85.1	100	99.4	80.6	77.4	94.2	63.7	84.9	85.9	88.9	72.0	73.0	66.5	100	91.1	86.8	87.3	100	79.6
Mask3D (Schult et al., 2023)	55.2	78.0	87.0	100	98.5	78.2	81.8	93.8	76.0	74.9	92.3	87.7	76.0	78.5	82.0	100	91.2	86.4	87.8	98.3	82.5
QueryFormer (Lu et al., 2023)	58.3	78.7	87.4	100	97.8	80.9	87.6	93.6	70.2	71.6	92.0	87.5	76.6	<u>77.2</u>	81.8	100	99.5	91.6	89.2	100	76.7
OneFormer3D (Kolodiaznyi et al., 2024a)	56.6	78.1	86.6	100	97.5	80.2	<u>87.7</u>	93.8	72.1	72.6	91.3	85.5	72.6	73.2	81.2	100	93.5	<u>92.4</u>	89.5	100	76.9
MSTA3D (Tran et al., 2024)	56.9	79.5	87.9	100	99.4	92.1	80.7	93.9	77.1	88.7	92.3	86.2	72.2	76.8	75.6	100	91.0	90.4	83.6	99.9	82.4
HieFormer (Ours)	58.5	80.1	88.6	100	99.6	93.5	88.2	92.1	78.3	86.6	<u>92.4</u>	86.0	73.5	74.8	76.4	100	91.2	90.1	87.2	100	83.2

Table 3

Performance comparisons of 3D instance segmentation on ScanNetv2 validation set in terms of mean average precision. Bold indicates the best option, and underline indicates the second-best option. † denotes results reproduced under our unified evaluation protocol, consistent with Table 6.

Methods	mAP	mAP ₅₀	mAP ₂₅
DyCo3D (He et al., 2021)	35.4	57.6	72.9
HAIS (Chen et al., 2021)	43.5	64.4	75.6
DKNet (Wu et al., 2022b)	50.8	66.7	76.9
SoftGroup (Vu et al., 2022)	46.0	67.6	78.9
PBNet (Zhao et al., 2023)	54.3	70.5	78.9
TD3D (Kolodiazhnyi et al., 2024b)	47.3	71.2	81.9
ISBNet (Ngo et al., 2023)	54.5	73.1	82.5
Mask3D (Schult et al., 2023)	55.2	73.7	83.5
SPFormer (Sun et al., 2023)	56.3	73.9	82.9
QueryFormer (Lu et al., 2023)	56.5	74.2	83.3
Sketchy-3DIS (Deng et al., 2025)	–	68.8	83.6
MAFT (Lai et al., 2023)	58.3	75.6	84.5
MSTA3D (Tran et al., 2024)	58.4	77.0	85.4
OneFormer3D† (Kolodiazhnyi et al., 2024a)	58.7	77.6	<u>85.9</u>
EASE (Roh et al., 2024)	60.2	77.2	–
HieFormer (Ours)	<u>59.3</u>	77.8	86.5

Table 4

Performance comparisons of 3D instance segmentation on ScanNet200 validation set. Bold indicates the best option, and underline indicates the second-best option.

Methods	mAP	mAP ₅₀	mAP ₂₅
PointGroup (Jiang et al., 2020)	–	24.5	–
PointGroup + LGround (Rozenberszki et al., 2022)	–	26.1	–
ISBNet (Ngo et al., 2023)	24.5	32.7	–
SPFormer (Sun et al., 2023)	25.2	33.8	39.6
TD3D (Kolodiazhnyi et al., 2024b)	23.1	34.8	<u>40.4</u>
MSTA3D (Tran et al., 2024)	<u>26.2</u>	<u>35.2</u>	40.1
HieFormer (Ours)	27.1	35.7	40.9

Table 5

Performance comparisons of 3D instance segmentation on Area 5 of the S3DIS dataset. Bold indicates the best option, and underline indicates the second-best option.

Methods	mAP ₅₀	mPrec	mRec
DyCo3D (He et al., 2021)	–	64.3	64.2
DKNet (Wu et al., 2022b)	–	70.8	65.3
HAIS (Chen et al., 2021)	–	71.1	65.0
SoftGroup (Vu et al., 2022)	66.1	73.6	66.6
TD3D (Kolodiazhnyi et al., 2024b)	65.1	74.4	64.8
PBNet (Zhao et al., 2023)	66.4	74.9	65.4
SPFormer (Sun et al., 2023)	66.8	72.8	67.1
MAFT (Lai et al., 2023)	69.1	–	–
QueryFormer (Lu et al., 2023)	69.9	70.5	72.2
Mask3D (Schult et al., 2023)	<u>71.9</u>	74.3	63.7
MSTA3D (Tran et al., 2024)	70.0	80.6	70.1
HieFormer (Ours)	72.5	81.2	<u>71.5</u>

Table 6

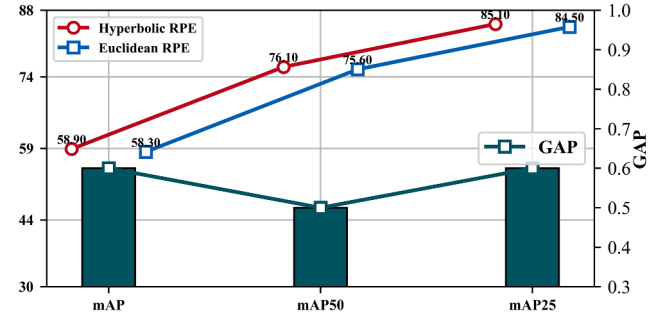
Results of 3D instance segmentation on ScanNetv2 validation set, including comparisons with representative transformer-based methods. The term ‘H’ represents the integration of HieFormer. † denotes results reproduced under our unified evaluation protocol.

Method	mAP	mAP ₅₀	mAP ₂₅
Mask3D† (Schult et al., 2023)	55.2	73.7	83.5
SPFormer (Sun et al., 2023)	56.3	73.9	82.9
MAFT (Lai et al., 2023)	58.3	75.6	84.5
OneFormer3D† (Kolodiazhnyi et al., 2024a)	58.7	77.6	85.9
H + Mask3D†	56.3 (+1.1)	74.2 (+0.5)	84.1 (+0.6)
H + SPFormer	59.1 (+2.8)	77.2 (+3.3)	85.3 (+2.4)
H + MAFT	59.6 (+1.3)	77.5 (+1.9)	85.6 (+1.1)
H + OneFormer3D†	59.3 (+0.6)	77.8 (+0.2)	86.5 (+0.6)

Table 7

Results of 3D instance segmentation on S3DIS Area 5, including comparisons with representative transformer-based methods. The term ‘H’ represents the integration of HieFormer. † denotes results reproduced under our unified evaluation protocol.

Method	mAP	mAP ₅₀	mAP ₂₅
Mask3D† (Schult et al., 2023)	57.8	71.9	77.2
SPFormer (Sun et al., 2023)	–	66.8	–
MAFT† (Lai et al., 2023)	47.1	62.8	71.2
OneFormer3D† (Kolodiazhnyi et al., 2024a)	58.7	72.0	–
H + Mask3D†	58.1	72.3	77.3
H + SPFormer	52.4	68.5	77.9
H + MAFT†	48.3	66.1	74.5
H + OneFormer3D†	59.6	72.5	79.1

**Fig. 8.** Ablation study on different relative position encodings.

tural separation. Excessively large curvature values ($c \geq 0.50$) degrade accuracy, because stronger metric expansion pushes features toward the Poincaré boundary and makes geodesic relations more sensitive. The learnable-curvature variant achieves the best overall performance, showing that jointly optimizing c with the network is preferable to imposing a manually selected fixed curvature for all training data.

Relative Position Encoding. The ablation study on position encoding strategies (Fig. 8) demonstrates that the proposed hyperbolic position encoding consistently outperforms its conventional Euclidean counterpart across all metrics, with improvements of +0.6% mAP (from 58.3% to 58.9%), +0.5% mAP₅₀ (from 75.6% to 76.1%), and +0.6% mAP₂₅ (from 84.5% to 85.1%). These results indicate that hyperbolic position encoding more effectively preserves hierarchical geometric relationships in 3D point clouds, particularly for capturing long-range dependencies and complex structural patterns. The consistent performance gains confirm that the proposed hyperbolic representation learning framework provides a more suitable positional context for 3D instance segmentation compared to traditional Euclidean approaches.

Attention Mechanisms. The ablation study on attention mechanisms (Table 8) demonstrates that the proposed hybrid attention consistently outperforms single-type attention across all metrics. Specifically, hybrid attention yields gains of +1.8% mAP (from 57.3% to 59.1%), +1.7% mAP₅₀ (from 75.5% to 77.2%), and +1.2% mAP₂₅ (from 84.1% to 85.3%). These improvements indicate that ECA and HPA serve complementary roles: ECA recalibrates channel-wise feature responses, while HPA enhances hierarchical spatial relational modeling. Their combination provides stronger feature discrimination and structural awareness, leading to superior performance compared to using either mechanism independently.

Ablation study of different modules. The comprehensive ablation study (Table 9) validates the synergistic effectiveness of our proposed HieFormer framework. The results demonstrate that each component contributes uniquely to the final performance, while the complete model, which combines Learnable Curvature (LC), Hyperbolic Relative Position Encoding (HRPE), and Hybrid Hyperbolic and Euclidean At-

Table 8

Ablation study on attention mechanisms.

Attention Method	mAP	mAP ₅₀	mAP ₂₅
SPFormer	56.3	73.9	82.9
ECA only	57.3	75.5	84.1
HPA only	58.2	75.2	84.6
Hybrid (HPA + ECA)	59.1	77.2	85.3

Table 9

Ablation study of different modules.

LC	HRPE	HHEA	mAP	mAP ₅₀	mAP ₂₅
			56.3	73.9	82.9
	✓	✓	57.9	75.8	84.6
✓		✓	56.7	74.9	84.4
✓	✓		56.1	74.2	83.9
✓	✓	✓	59.1	77.2	85.3

Table 10

Inference time ablation of different geometric modules in HieFormer on the ScanNetv2 validation set. All results are measured on one RTX 3090 GPU.

Method Variant	Time (s)	Param. (M)
Baseline (Euclidean only)	0.72	18.23
+ LC	0.74	18.23
+ LC + HRPE	0.76	18.23
+ LC + HRPE + HPA	0.85	20.32
+ LC + HRPE + HPA + ECA	0.91	21.06

tention (HHEA), achieves the best performance. Notably, the complete model outperforms all partial combinations. These findings confirm that learnable curvature provides adaptive geometric representation capabilities; hyperbolic position encoding effectively captures hierarchical spatial relationships; hybrid attention enables comprehensive feature aggregation. The consistent performance gains across all metrics demonstrate that our framework's components work synergistically to achieve state-of-the-art 3D instance segmentation performance, with each module contributing distinct and complementary advantages to the overall system.

Inference Time Analysis. Table 10 reports the inference-time ablation of the proposed geometric components. The Euclidean-only baseline achieves the fastest runtime (0.72s) with 18M parameters. Adding the LC module introduces almost no computational overhead (+0.02s), and incorporating HRPE increases the cost only slightly (+0.02s), demonstrating that local hyperbolic encoding remains lightweight. The Hyperbolic Positional Attention (HPA) contributes a more noticeable increase (+0.09s) due to global curvature-aware aggregation, while keeping parameters modest at 20M. The full model with ECA reaches 0.91s and 21M parameters, representing the complete geometric hierarchy. Overall, each geometric component adds meaningful representational capacity with only moderate increases in runtime and parameters. The full configuration remains efficient on a single RTX 3090 GPU, confirming that the integration of hyperbolic and Euclidean reasoning enhances segmentation performance without sacrificing practical inference speed.

Multi-seed Robustness Analysis. Table 11 reports the robustness of different model variants on the ScanNetv2 validation set across three random seeds. All configurations exhibit very small standard deviations, indicating stable optimization behavior and limited sensitivity to random initialization. In particular, the baseline with LC + HRPE achieves 56.0 ± 0.1 mAP, while replacing HRPE with HHEA yields a modest but consistent improvement to 56.4 ± 0.3 mAP, suggesting that hybrid attention alone provides complementary benefits beyond positional encoding. Notably, combining both HRPE and HHEA results in a clear performance gain, achieving 58.8 ± 0.4 mAP. The improvement of ap-

Table 11Multi-seed robustness analysis on ScanNetv2 validation set. Results are reported as mean $\pm$ standard deviation over three random seeds.

Baseline	mAP (%)
+ LC + HRPE	56.0 ± 0.1
+ LC + HHEA	56.4 ± 0.3
+ LC + HRPE + HHEA (Ours)	58.8 ± 0.4

Table 12

Sensitivity analysis to superpoint granularity on ScanNetv2.

Superpoints	mAP	mAP ₅₀	mAP ₂₅
100 (coarse)	56.8	74.2	83.1
200	57.6	75.0	83.9
300	58.4	76.1	84.6
400 (default)	59.1	77.2	85.3
500	58.6	76.9	84.8
600	57.9	75.6	84.5
800 (fine)	57.2	74.4	84.3

proximately +2.8 mAP over LC + HRPE is substantially larger than the observed run-to-run variability, suggesting that the gain is robust to random initialization. This result highlights the complementary nature of hyperbolic relative positional encoding and hybrid hyperbolic and Euclidean attention, and supports the effectiveness of jointly modeling hierarchical structure and feature interactions. For clarity, the single-run numbers reported in the ablation tables (e.g., Tables 8 and 9) correspond to the best-performing of the three seeds, whereas Table 11 reports the mean and standard deviation over the same three seeds; the two are therefore mutually consistent within the reported variance.

Superpoint Granularity Sensitivity Analysis. Table 12 reports the impact of superpoint granularity on ScanNetv2. As the number of superpoints increases from 100 to 400, all evaluation metrics (mAP, mAP₅₀, and mAP₂₅) improve steadily, with mAP rising from 56.8 to 59.1. This trend indicates that overly coarse superpoints tend to merge multiple semantic or instance-level structures, limiting the model's ability to resolve fine-grained instance boundaries and hierarchical relationships. The best performance is achieved at 400 superpoints, which suggests a favorable balance between structural abstraction and spatial detail. At this granularity, superpoints are sufficiently fine to preserve instance-level geometry, while still maintaining stable and semantically meaningful regions for hierarchical modeling in hyperbolic space. This aligns well with the design motivation of HieFormer, which relies on superpoints as intermediate structural units to capture hierarchical relationships rather than raw point-level noise. However, further increasing the number of superpoints beyond 400 leads to a gradual performance degradation. When superpoints become too fine-grained (e.g., 600 or 800), they tend to fragment coherent object parts into small regions, weakening the structural consistency of the hierarchy and introducing additional noise into query-superpoint associations. This effect is particularly visible in mAP₅₀ and mAP₂₅, where the gains diminish or slightly drop, indicating reduced robustness under stricter overlap thresholds.

Failure Case Analysis. Fig. 9 presents representative failure cases of our method. In the left example, the model merges two adjacent objects into a single instance, despite their distinct ground-truth labels. Both objects exhibit highly similar planar surfaces and almost indistinguishable color cues. Since hyperbolic embeddings emphasize hierarchical structural differences, the lack of strong geometric boundaries causes both instances to collapse into a nearby region of the manifold, leading to under-segmentation. The middle example illustrates an over-segmentation failure in a cluttered indoor region. Here, multiple small objects are densely packed, and slight variations in local geometry are

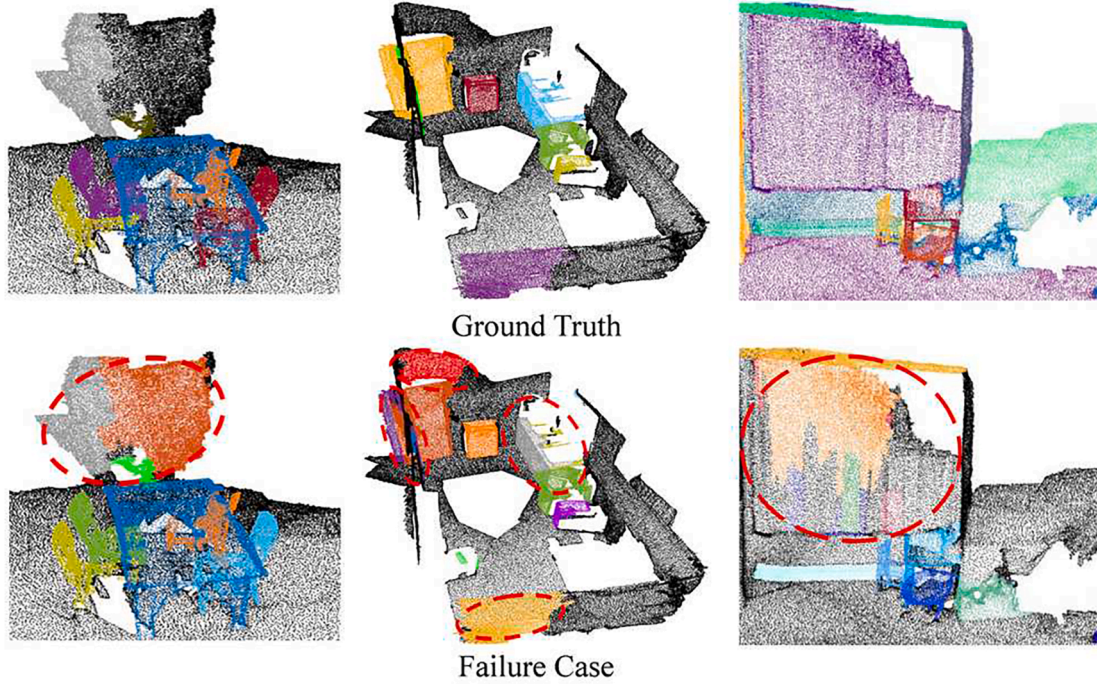


Fig. 9. Failure case visualization on ScanNetv2. Top: ground-truth instance annotations. Bottom: predictions produced by HieFormer. (Left) Under-segmentation occurs when two adjacent furniture pieces with weak geometric boundaries are merged into a single instance. (Middle) Over-segmentation appears in cluttered regions where densely packed objects cause excessive splitting of continuous surfaces. (Right) Boundary distortion arises when points near the Poincaré-ball boundary suffer from amplified hyperbolic metric expansion, leading to unstable and fragmented masks. Red dashed circles highlight the failure regions.

exaggerated by the hyperbolic positional attention. As a result, surfaces that should belong to the same instance are incorrectly separated into multiple predictions, producing fragmented masks. In the right example, the predicted mask exhibits clear distortions along the object boundary. Points lying close to the edge of the Poincaré ball experience amplified metric expansion, and even minor positional noise becomes significantly magnified in hyperbolic space. This leads to unstable query updates in the decoder and results in irregular or incomplete segmentation of the object. These failure cases reveal that while hyperbolic geometry substantially enhances hierarchical reasoning and instance discrimination, its effectiveness depends on the availability of reliable geometric cues, and extreme structural ambiguity or densely packed clutter can still challenge the model.

From a technical perspective, the boundary distortion issue stems from the conformal factor $\lambda_x = \frac{2}{1-c\|x\|^2}$, which increases rapidly as points approach the manifold boundary ($\|x\| \rightarrow 1/\sqrt{c}$), amplifying small perturbations in point coordinates. For under-segmentation, when the hyperbolic distance between instance centroids is small relative to the intra-instance spread, the hierarchical separation may not sufficiently overcome local feature ambiguity. Several promising directions could mitigate these issues in future work: (1) introducing boundary-aware regularization through an auxiliary loss that explicitly supervises normal discontinuities or geometric edges to assist the hyperbolic embeddings in separating adjacent instances; (2) applying adaptive curvature scaling that dynamically adjusts the curvature c or the embedding norm based on local point density, preventing features from prematurely converging to the ball boundary in crowded scenes; and (3) employing a hybrid distance metric that combines the hyperbolic distance for hierarchical structure with a local Euclidean-based boundary penalty to ensure both global organization and local precision.

4.6. Representation analysis under structural incompleteness

The segmentation and ablation results above establish the task-level effectiveness of HieFormer, but they do not show how the learned repre-

sentations respond when structural evidence is progressively removed. We therefore perform a paired representation analysis on the full ScanNetv2 validation split. For each of the 312 scenes, we randomly retain 100%, 80%, and 50% of the instance points during feature extraction. For each retention level, we record the 256-dimensional query representation at the corresponding position in the final Transformer decoder layer, using the same extraction position for the Euclidean baseline and HieFormer. Each retention condition therefore contains 312 samples, giving 936 representations per model. The three conditions preserve the identity of each scene while progressively reducing the structural information available to the decoder. This controlled setting allows us to examine whether the representation changes in an ordered manner as the observation becomes less complete.

Paired representation response analysis. Let $\mathbf{f}_s^r \in \mathbb{R}^{256}$ denote the final decoder representation for scene s at retention level $r \in \{100, 80, 50\}$. For each model, we estimate the per-dimension standard deviation from the 312 representations at 100% retention. Standard deviations smaller than 10^{-8} are replaced by the median positive standard deviation, giving the scale vector $\tilde{\sigma}$. The displacement from the corresponding full observation is then defined as

$$D_r(s) = \frac{1}{\sqrt{256}} \left\| \frac{\mathbf{f}_s^r - \mathbf{f}_s^{100}}{\tilde{\sigma}} \right\|_2, \quad r \in \{80, 50\}, \quad (22)$$

where the division is applied element-wise. We then define the monotonicity margin as $M(s) = D_{50}(s) - D_{80}(s)$. A positive margin indicates that the stronger 50% retention perturbation moves the representation farther from its own full-observation reference than the milder 80% perturbation. Fig. 10 shows the distribution of this margin over all 312 scenes. HieFormer exhibits the expected ordering in 273 scenes, corresponding to 87.5%, whereas the Euclidean baseline does so in 188 scenes, or 60.3%. We do not interpret a larger absolute displacement as inherently better. The relevant observation is the consistency of the ordering within the same scene and model. The substantially higher monotonicity rate of HieFormer indicates a more organized response to

Representation response to progressive structural incompleteness

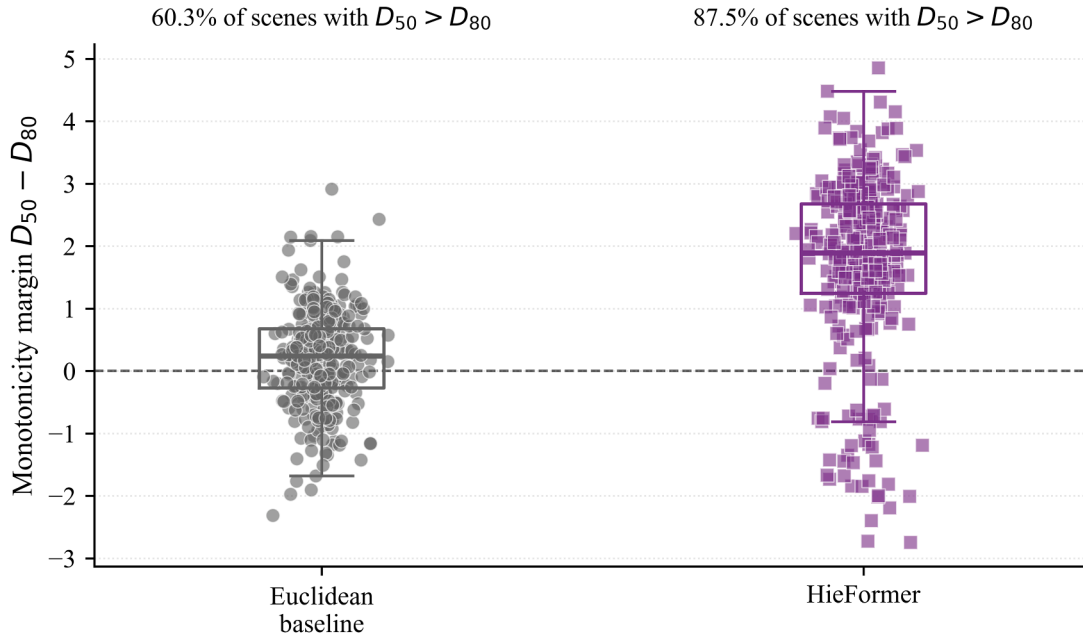


Fig. 10. Quantitative analysis of final decoder query representations under progressive structural incompleteness. For each ScanNetv2 validation scene, D_{80} and D_{50} measure the standardized 256-dimensional feature displacement from the corresponding 100% retention representation after retaining 80% and 50% of instance points. We report the monotonicity margin $M = D_{50} - D_{80}$, where $M > 0$ indicates a larger representation change under stronger incompleteness. HieFormer satisfies this ordering in 87.5% of scenes, compared with 60.3% for the Euclidean baseline. Points denote individual scenes and box plots summarize the distributions.

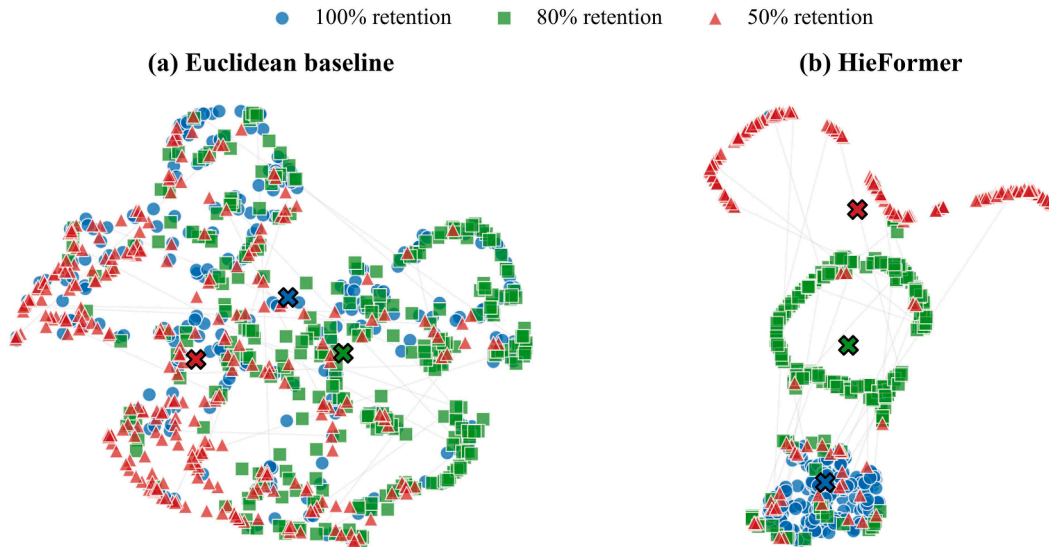


Fig. 11. UMAP visualization of the same final decoder query representations under 100%, 80%, and 50% instance point retention. Each model is embedded separately with the same UMAP configuration: 25 neighbors, minimum distance 0.1, Euclidean metric, and random seed 42. Only within-panel organization is interpreted. All 936 samples per model are shown. Light gray lines trace 12 fixed scene trajectories across retention levels, and the large X markers denote condition centroids. HieFormer exhibits clearer condition-dependent organization as structural observations are reduced, whereas the Euclidean baseline remains more intermixed. UMAP is used only for qualitative visualization.

progressively reduced structural observations in the final decoder representation.

Qualitative representation visualization. Fig. 11 visualizes the same 936 representations for each model using UMAP. The Euclidean baseline and HieFormer are embedded separately with identical UMAP settings: 25 neighbors, minimum distance 0.1, Euclidean metric, and random seed 42. All samples are shown, and 12 fixed scene trajectories

are included only to make the paired progression across 100%, 80%, and 50% retention easier to inspect. The large X markers indicate the centroid of each retention condition. The Euclidean baseline shows substantial overlap among the three conditions, whereas HieFormer exhibits a clearer organization as structural information is progressively removed. Because UMAP is nonlinear and fitted separately for the two models, we interpret only the organization within each panel and do not

compare absolute coordinates across panels. This visualization complements the quantitative result in Fig. 10. The two analyses therefore provide representation-level evidence that is consistent with the hierarchy-aware structural modeling intended by HieFormer, without treating the perturbation protocol as a ground-truth semantic hierarchy.

4.7. Limitations

Despite its strong performance, HieFormer has several limitations. First, the current implementation relies on pre-computed superpoints for query initialization and feature aggregation, which may limit applicability when high-quality superpoints are unavailable. Second, the current curvature parameter is learned globally and shared across scenes and instances. This design provides a stable common manifold and low overhead, but it cannot adapt the curvature to scene- or instance-specific structural complexity at inference time. A scene-conditioned or instance-conditioned curvature predictor is therefore a promising extension. Third, operating in hyperbolic space introduces numerical constraints such as norm clipping near the Poincaré-ball boundary and consequently requires careful optimization. Fourth, the representation analysis in Section 4.6 uses controlled random removal of instance points to reduce the structural evidence available to the decoder. This protocol directly probes representation behavior under incompleteness, but it does not evaluate recovery of a ground-truth part-whole hierarchy because the employed benchmarks do not provide exhaustive parent-child annotations for all scene components. Future work could evaluate explicit hierarchical labels or tree-distance correlations on datasets with richer structural annotations. Finally, although HieFormer achieves state-of-the-art or highly competitive results on indoor datasets, its generalization to large-scale outdoor scenes with unbounded spatial extent and dynamic objects remains to be established.

5. Conclusion

We presented HieFormer, a dual-geometry Transformer for 3D instance segmentation that combines hyperbolic and Euclidean representations within a unified decoder. Learnable curvature and HRPE provide hierarchy-aware geometric and positional modeling in hyperbolic space, while HHEA couples this structural representation with Euclidean feature interactions for fine-grained instance discrimination. Experiments on ScanNetv2, ScanNet200, and S3DIS show state-of-the-art or competitive performance across multiple backbone architectures, and the ablation studies demonstrate the complementary contributions of the proposed components. The representation analysis further shows that HieFormer responds more consistently to progressive loss of structural observations than the Euclidean baseline, while the UMAP visualization reveals clearer condition-dependent organization in the final decoder features. These results provide complementary evidence for the intended hierarchy-aware structural modeling without relying only on task-level metrics. Future work will explore scene-conditioned or instance-conditioned curvature, explicit part-whole supervision, and extensions to large-scale outdoor 3D scenes.

CRediT authorship contribution statement

Xiaoyu Wei: Writing – original draft, Visualization, Software, Methodology, Formal analysis, Conceptualization; **Anton Konushin:** Writing – review & editing, Validation, Supervision, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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